INFORMATION TO KEEP WITH YOU

My complete treatment handbook

Lutetium-177 oxodotreotide (Lutathera / peptide receptor radionuclide therapy (PRRT))

Understand your treatment. Know what to expect. Keep your plan to hand.

Planned treatments
4
Specialist PRRT administrations

Usual spacing
8 weeks
Between administrations

Regimen source
NCCP 00642
Version 2

Your planned course

Four planned administrations, usually 8 weeks apart; each needs specialist clearance.

1	2	3	4
Treatment 1	Treatment 2	Treatment 3	Treatment 4
Week 0	Week 8	Week 16	Week 24

Illustrative elapsed weeks from the first treatment. Each administration needs specialist clearance. Dates can change; follow-up continues after treatment 4.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Seek urgent help for sudden severe flushing with faintness, severe diarrhoea, wheeze or breathing trouble, especially in the first 48 hours; follow the hormone-crisis plan. Fever, bleeding, much less urine, persistent severe abdominal pain, muscle weakness or confusion needs prompt advice. Tell emergency staff when you received PRRT; radiation precautions must not delay urgent care.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

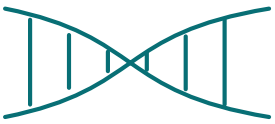
Targeted radiation and kidney protection

PRRT carries radiation to somatostatin receptors on neuroendocrine tumour (NET) cells to help control disease that cannot be removed or has spread. A specialist NET and nuclear-medicine team checks receptor imaging, organ function and your priorities before recommending it.

RECEPTOR-TARGETED RADIOPHARMACEUTICAL

Lutetium-177 oxodotreotide

Lutathera / 177Lu-DOTATATE / PRRT



Carries radiation to somatostatin receptor targets

Lutathera joins a receptor-binding molecule to the radioactive element lutetium-177. After infusion, it attaches to somatostatin receptors on suitable NET cells and delivers radiation nearby. The receptor scan helps select patients. Kidneys and blood-forming marrow can also receive radiation, which is why kidney protection and long-term blood monitoring matter.

somatostatin receptor (SSTR) target on NET cells → Radiolabelled medicine attaches → Radiation damages tumour cells

What to watch for

Keep all between-treatment and after-treatment blood tests; use your personalised radiation-safety and hormone-symptom plans.

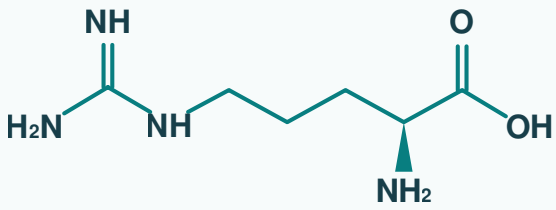
SUPPORTIVE INFUSION

Lysine / arginine amino acids

Kidney-protective infusion



L-lysine



L-arginine

Reduces kidney uptake of radioactivity

The amino-acid infusion reduces the amount of radioactive medicine retained in the kidneys. It is an essential part of kidney protection. It can cause nausea and affect potassium or fluid balance, so anti-sickness treatment, blood tests and observation are needed.

Amino acids infused → Kidney uptake reduced → Kidney radiation reduced

What to watch for

Tell staff promptly about nausea, palpitations, weakness, breathlessness or swelling during infusion.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Understanding your particular NET

Several different features help the team choose treatment.

Where it started and where it is now

A neuroendocrine tumour begins in cells with hormone-making features. Its starting organ and the places it has spread are separate parts of the diagnosis. A pancreatic neuroendocrine tumour differs from pancreatic adenocarcinoma; a deposit in the liver may have come from another organ.

Differentiation, grade and Ki-67

The pathology report describes how organised the cells look (differentiation) and their growth activity (grade). Ki-67 measures cells showing a growth-related protein in the sample. Its percentage describes that sample, not the amount of your body affected or your chance of survival.

Grade is different from stage

Stage describes extent; grade describes tumour biology. Either can vary independently. A well-differentiated grade 3 neuroendocrine tumour and a poorly differentiated neuroendocrine carcinoma are distinct diagnoses and may need different treatments.

Functioning and nonfunctioning

A functioning tumour produces a hormone syndrome. A nonfunctioning tumour can still cause problems through growth or pressure. Somatostatin receptor uptake on a scan shows a treatment target; it does not, on its own, identify a hormone syndrome.

Your diagnosis record

Ask for one diagnosis summary: starting organ, grade, differentiation, Ki-67 result if reported, sites involved, hormone syndrome and receptor-imaging result. Keep pending results clearly marked.

Your overall plan and choices

Your priorities belong in the treatment plan.

What are we trying to achieve?

Agree your treatment aim: slowing growth, reducing tumour size, controlling hormone symptoms or preserving daily function. A scan showing no growth may be a useful result even when the tumour has not shrunk.

How treatments fit together

Your plan may use different treatments at different times, including surgery, medicines, receptor-targeted radiation or treatment directed at the liver. Ask the team to mark your own sequence and who coordinates it.

Choosing between options

Ask why the proposed option fits the tumour site, grade, receptor findings, symptoms and your health. Discuss reasonable alternatives, observation when appropriate and suitable trials. Your team explains which options are available to you.

Evidence is about groups

Studies measure different outcomes, such as shrinkage, time before worsening and survival. Ask which outcome matters for your plan and how closely the study group matches your diagnosis. No study average can predict one person’s course.

Review points and shared decisions

Agree when to review the plan and what could lead to continuing, pausing or changing treatment. Include the effects on work, caring duties, travel and the activities that matter to you.

What NETTER-1 did and did not show

The trial studied progressive SSTR-positive midgut NET after octreotide treatment.

Lutetium-177 DOTATATE with standard-dose octreotide improved time before progression compared with high-dose octreotide.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

Preparation, treatment day and follow-up

1 Treatment 1 **Week 0** — 2 Treatment 2 **Week 8** — 3 Treatment 3 **Week 16** — 4 Treatment 4 **Week 24**

Illustrative spacing. Each administration needs specialist clearance; dates can change.

T1–T4

Four planned PRRT administrations

T1 T2 T3 T4

Usually eight weeks between administrations, with blood tests every two weeks and a PRRT clinic review. The service confirms each date and any delay.

Four administrations are intended, not guaranteed. Blood results, recovery, specialist review and the service plan determine whether and when each can be given.

BASELINE → TREATMENT → FOLLOW-UP

PRRT tests before, between and after four intended administrations

This is the NCCP blood-test sequence with EMA potassium safety and the PRRT service pathway. The post-infusion radiation-assessment scan is different from a response scan. The service confirms bookings, exact dates and any additional follow-up.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00642 v2 · Lutathera regimen · EMA Lutathera product information and patient leaflet · updated 14 August 2026 · HSE / PRRT service: introductory patient information

Tests in your plan

Full blood count

WHEN

Before treatment; once 2–4 weeks before and shortly before each administration; every 2 weeks between treatments.

Checks haemoglobin, white cells with differential, and platelets.

After the last administration: every 4 weeks for at least 3 months, then every 3 months for the following year. Your written plan may include extra early checks.

Kidney function

WHEN

With the blood-test schedule above; monitor for at least the first year after treatment.

Serum creatinine and calculated creatinine clearance assess kidney safety.

The team calculates clearance using the required method. A normal symptom check does not replace these tests.

Liver function

WHEN

With the blood-test schedule above.

Tests include ALT, AST, albumin, bilirubin and INR to assess liver function and protein production.

Potassium

WHEN

Before the kidney-protective amino-acid infusion; further checks if indicated.

Detects potassium disturbances that the amino-acid infusion can worsen.

Heart pumping assessment

WHEN

Before starting, as specified in the regimen.

An echocardiogram or MUGA scan assesses heart pumping function.

5-hydroxyindoleacetic acid (5-HIAA)

If part of your plan

WHEN

Only if requested for your hormone syndrome; the team sets the dates.

Measures a breakdown product of serotonin to help assess selected neuroendocrine tumours.

Often a timed urine collection. Follow the laboratory's food and medicine instructions.

Chromogranin A

If part of your plan

WHEN

Only when the team considers it informative for your tumour.

A selected tumour-marker result, interpreted with symptoms and scans.

Acid-suppressing medicines and other conditions can alter the result. It is not a routine test before every treatment; do not stop medicines yourself.

Selected pancreatic hormones

If part of your plan

WHEN

Only for the hormone question identified by your specialist.

Examples include gastrin, insulin or glucagon when the diagnosis or symptoms call for them.

The team names the test, preparation and review date; these are not a standard panel for every patient.

THE WHOLE TREATMENT JOURNEY

Lutetium-177 oxodotreotide (Lutathera / PRRT): treatment and checks

Progressive, unresectable or metastatic, well-differentiated G1/G2 somatostatin-receptor-positive gastroenteropancreatic NET under NCCP 00642a.

Four administrations are planned, usually eight weeks apart. Lutathera is infused over about 30–40 minutes, with a four-hour kidney-protective amino-acid infusion and anti-sickness treatment. The visit is much longer than the Lutathera infusion; an overnight stay may be needed.

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before the course	Before each administration	Between administrations	After the final dose	NET response / safety plan
Blood count / kidneys / liver NCCP 00642	Baseline tests	Once 2–4 weeks beforehand AND shortly before infusion	Every 2 weeks	Every 4 weeks for at least 3 months, then every 3 months for the following year	Additional tests or longer follow-up if abnormal
Potassium / renal clearance NCCP 00642	Assess renal function and electrolyte risks	Potassium before amino acids; renal function and suitability reviewed	Renal checks during therapy; earlier assessment if ill	Renal monitoring for at least the first year	Fluid and electrolyte plan individualised
Heart / pregnancy / hormones NCCP 00642	echocardiogram (heart ultrasound) (ECHO) or MUGA; pregnancy testing where applicable; hormone plan	Review symptoms and any pregnancy possibility; somatostatin analogue (SSA) plan confirmed	Report hormone symptoms, especially during first 48 hours	Follow hormone symptoms and contraception plan	Additional heart tests if clinically indicated

Care / check	Before the course	Before each administration	Between administrations	After the final dose	NET response / safety plan
Imaging / clinic NCCP 00642	SSTR-positive disease and baseline assessment	Specialist clearance for each treatment	PRRT clinic review; special scan(s) after each infusion for radiation assessment	Response and follow-up imaging as agreed	No universal mandatory tumour-marker or response positron emission tomography (PET) interval
Radiation precautions NCCP 00642	Discuss household, travel, work and care needs	Receive written instructions for this administration	Observe each restriction until its own end date	Keep discharge information and tell other treating teams	Medical physicist confirms personal restrictions

What each check is for

1 Four planned PRRT administrations

T1–T4

PRRT carries radiation to somatostatin receptors on NET cells to help control disease that cannot be removed or has spread. A specialist NET and nuclear-medicine team checks receptor imaging, organ function and your priorities before recommending it.

Ask / act: Confirm the current prescription, tests and support medicines.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Practical treatment planning

Agree the arrangements with your oncology team.

STEP 1

Before admission

Confirm SSTR imaging, appointments, recent bloods, hormone medicines and pregnancy precautions. Tell the team about kidney/heart problems, incontinence, a catheter, childcare or close contact with pregnant people; these may change the practical plan.

STEP 2

Treatment day

The standard NCCP activity is 7.4 GBq by intravenous infusion, usually every eight weeks for four administrations; the specialist may alter, delay or stop treatment. Lutathera runs over approximately 30–40 minutes. A lysine/arginine amino-acid infusion starts 30 minutes beforehand and continues for four hours. An anti-sickness medicine is given at least 30 minutes before the amino acids. The first visit generally includes an overnight stay; later treatments may be day case or overnight. The service confirms your plan.

STEP 3

SSA injections: stop and restart

The NET and nuclear-medicine teams must coordinate the injection plan. Do not translate “about four weeks” into your own date: NCCP specifies 30 days for long-acting SSA and 24 hours for short-acting SSA before PRRT. Restart timing follows the written service plan; it can differ with day-case arrangements.

STEP 4

At home after treatment

Before discharge, obtain a paper plan with the end date for each restriction on contact, sleeping, work, travel and body-fluid handling. First-week instructions include daily bathing, careful handwashing, sitting to pass urine and separately washing body-fluid-soiled laundry. Ask for specific advice about pads, incontinence or a catheter. Examples include sleeping separately from a partner for 14 days and from pregnant people or children for three weeks; restrictions differ by contact and circumstance. Your medical physicist sets your personal instructions. Tell them before discharge if you cannot follow the plan at home.

STEP 5

Travel and other healthcare

Carry the discharge letter and the official radiation-treatment card for the period the medical physicist specifies. They identify the treatment and date and help with travel, radiation detectors and other healthcare. The SACTCheck wallet card is an additional record and does not replace them. Tell healthcare staff about the treatment; radiation precautions must never delay urgent care.

STEP 6

Pregnancy, breastfeeding and fertility

Lutathera can harm an unborn child and may affect fertility. Discuss fertility preservation before treatment. Use effective contraception during treatment and for seven months after the last dose for patients who could become pregnant; men with partners who could become pregnant need contraception during treatment and for four months afterwards. Breastfeeding must stop before treatment; follow specialist advice.

STEP 7

If a visit is delayed

Low blood counts, kidney or liver changes, other side effects or a change in the cancer may lead the specialist to postpone, reduce or stop treatment. A four-stamp record shows planned administrations, not medical clearance. A dose does not automatically become suitable at the maximum interval. Keep the blood-test and symptom plan while waiting for the team's decision.

STEP 8

Contacts and paperwork

Write the actual NET, PRRT nursing and radiation-protection contacts on the paper plan. Example numbers are not clinical contact lines. Keep the written precautions and urgent hormone-symptom instructions with the discharge letter.

STEP 9

Your specialist team's scan appointments

Radiation-assessment scans measure where the treatment went and the radiation delivered to organs. They answer a different question from computed tomography (CT) or magnetic resonance imaging (MRI) scans of tumour response. The detailed treatment information describes three radiation scans on separate days in the first week after the first infusion, starting the next day, and one after later infusions; response CT or MRI is described between the second and third infusions and after the course. Confirm your actual bookings with the team: these are service arrangements, not a universal timetable for every neuroendocrine tumour.

STEP 10

Extra blood tests after the course

Blood counts, kidney and liver tests continue every four weeks for at least three months after the last administration, then every three months for the following year. Your written discharge plan may also include two-weekly tests for the first eight weeks after the last infusion. The team confirms those extra early appointments and who reviews the results.

Later effects and continuing follow-up

New or persistent symptoms still need review after the course.

Delayed effects

Kidney injury and rare bone-marrow disorders, including myelodysplastic syndrome or acute leukaemia, can occur later. Persistent tiredness, infections, bruising or bleeding needs review even after the treatment course.

Making sense of tests and results

Safety tests, tumour tests and hormone tests answer different questions.

Blood tests for treatment safety

Blood counts assess oxygen-carrying cells, infection-fighting cells and platelets. Kidney and liver results help assess treatment safety. These checks are needed even when you feel well; a reassuring scan answers a different question.



CT, MRI and receptor imaging

CT and MRI show anatomy and tumour size. Somatostatin receptor imaging tests whether a particular receptor target is present. An fluorodeoxyglucose positron emission tomography (FDG-PET) scan measures uptake of a glucose-like tracer and is used selectively. Your team chooses the scans that can guide your care.

Hormones and tumour markers are individual

Selected markers may help answer a hormone or tumour question. Examples include 5-hydroxyindoleacetic acid (5-HIAA) for serotonin-related symptoms, chromogranin A, or a specific pancreatic hormone. They are not a routine panel for every neuroendocrine tumour. Interpret changes with symptoms and imaging.

Preparing for a test

Use the laboratory instructions for fasting, food restrictions and any timed urine collection. Some medicines affect marker results, including acid-suppressing medicines and chromogranin A. Ask the team what to do; do not stop a prescribed medicine yourself.

Heart and inherited-risk checks

Some hormone syndromes can affect heart valves, so an ultrasound of the heart may be requested. Inherited-risk assessment depends on the tumour and family history; multiple endocrine neoplasia type 1 is one example. An inherited-risk test and a test of tumour tissue answer different questions.

Who does what, and when?

For each test, write its name, purpose, booking date and result-review contact. Ask the team to set your next scan or marker date. A schedule for someone after surgery may not apply during your treatment.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

Day 1 → Day 56: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

Treatment or check	Timing in this view
PRRT and protective infusions	 **Day 1** Anti-sickness treatment, amino acids and Lutathera; admission / overnight plan confirmed.
Personal radiation-safety plan	 **Days 1–21** Restriction periods differ by activity and contact; follow your written end dates, not the band alone.
Fortnightly blood tests and PRRT review	 **Days 1–56** Blood count, kidneys and liver; dates are booked individually.
Assessment for next administration	 **Day 56** Usually eight weeks later; proceed only after specialist approval.

Four administrations, not four consecutive days: The intended administrations are usually eight weeks apart. Record each actual date and any change; delays do not authorise a catch-up treatment.

Follow-up extends beyond the fourth treatment: Blood counts, kidneys and liver require continued checks. Kidney injury and rare marrow disorders may appear later, so retain the follow-up schedule and report new symptoms.

Two kinds of scan: Your specialist team plans special scans after each infusion to measure radiation delivered to organs. computed tomography (CT)/magnetic resonance imaging (MRI) or other scans to assess tumour response follow the specialist plan.

Radiation safety has individual end dates: A contact restriction, work restriction and sleeping arrangement may finish on different dates. The wallet or calendar is a reminder; your written radiation-safety instructions control the details.

> **Why these bands stay broad:** The 56-day view shows a usual interval, not a guarantee of the next administration. The radiation band shows a possible period only; each restriction has its own written end date from the service. Symptoms and abnormal tests can persist after any band ends. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00642 v2 · Lutathera regimen ↗ · EMA Lutathera product information and patient leaflet · updated 14 August 2026 ↗ · HSE / PRRT service: introductory patient information ↗

There are no guaranteed "safe days". Follow your oncology unit's emergency instructions. Fever, shivering or suddenly feeling unwell needs urgent advice; do not wait for the next appointment. Call emergency services for collapse, severe breathing difficulty, chest pain or uncontrolled bleeding.

Between-treatment notes

Day 1–2: Treatment and early observation

Plan a long hospital visit; the first treatment generally involves an overnight stay. Staff give anti-sickness medicine and kidney-protective amino acids. Hormone reactions are especially important in the first 48 hours. Before discharge, obtain written emergency contacts, SSA timing and radiation precautions.

Day 3–21: Recovery and individual precautions

Energy, appetite and bowel symptoms may change. Radiation precautions may last two to three weeks for some activities or contacts; exact restrictions come from the medical physicist. Attend blood tests every two weeks. Call for fever, bleeding, reduced urine, severe pain or sudden hormone symptoms.

Day 22–56: Tests and next-treatment decision

Continue fortnightly bloods and attend the PRRT review. A post-infusion scan checks radiation distribution; response scans follow a separate specialist plan. The team confirms whether and when the next administration is suitable. Do not treat an eight-week date as automatic clearance.

Support around the medicines

1 Kidney protection

A lysine/arginine infusion is required for kidney protection and begins before Lutathera. Follow the service's fluid and toileting advice. Tell staff about heart or kidney problems; do not use a generic fluid target.

2 Nausea prevention

PRRT has moderate emetogenic potential. A 5-HT3 anti-sickness medicine is given at least 30 minutes before the amino-acid infusion, with additional treatment if needed. Tell staff promptly if nausea starts.

3 Somatostatin analogue coordination

Long-acting SSA injections can interfere with receptor targeting. NCCP uses a 30-day interval before PRRT and short-acting injections are withheld for 24 hours. The referring NET consultant decides your exact stop and restart plan, balancing hormone control. Never stop injections on your own.

4 Radiation safety and practical planning

The medical physicist gives individual written and verbal advice. Contact with young children or pregnant people may need restriction for up to three weeks, and close-contact work may require up to two weeks away. Arrange transport, sleeping, toileting, laundry and caring responsibilities before admission.

Hormone and fluid changes to watch for

Use the urgent-contact instructions whenever symptoms occur.

Hormone crisis

PRRT can trigger a sudden hormone release. The treatment information highlights the first 48 hours as the main early-risk period. Staff monitor you and may give extra medicines or extend the stay. You need clear instructions for symptoms after discharge; a history of hormone symptoms affects planning.

Tumour breakdown and fluid changes

Rapid tumour breakdown can disturb blood chemistry. New muscle cramps, weakness, confusion, palpitations, reduced urine or breathlessness needs prompt assessment. Staff also monitor potassium and fluid balance around the amino-acid infusion.

Hormone symptoms and day-to-day changes

A new symptom may come from the NET, treatment or another illness.

Build a picture of what is normal for you

Keep a brief record of your usual bowel pattern, flushing, appetite, weight, pain and energy. Note new changes and when they started. This helps the team assess whether treatment, hormones or another problem needs attention.

Hormone-related symptoms

Hormone symptoms depend on the tumour: examples include flushing and watery diarrhoea, acid-related symptoms, or changes in blood sugar. An insulin-producing tumour can lower blood sugar. Ask which symptoms apply to you and whether you need a written rescue plan.

Hormone treatment is a separate part of the plan

Hormone-control medicines have their own schedule. A pause in anticancer tablets does not automatically mean pausing those medicines. Before peptide receptor radionuclide therapy, the specialist team coordinates any interruption and restart of somatostatin analogue injections.

When a change is urgent

Follow your oncology emergency plan for sudden or severe symptoms. Collapse, severe breathing difficulty, chest pain, confusion or uncontrolled bleeding needs emergency assessment. Severe diarrhoea, persistent vomiting, dehydration, fever or sudden illness needs urgent oncology advice. Do not wait for a scan, an online answer or the next clinic visit.

Procedures and hormone syndromes

Tell teams arranging an operation, anaesthetic or other procedure about a hormone-producing tumour or previous hormone crisis. Your specialist decides whether extra preparation is needed. Carry the agreed emergency instructions and current medicine list.

Eating, energy, practical help and support

Supportive care can be used alongside treatment from the start.

Food, weight and digestion

Report weight loss, persistent diarrhoea, greasy stools or difficulty eating. Different causes need different treatment. Ask for a dietitian and whether reduced digestive-enzyme function needs assessment.

Fluids and safe activity

Use your personal fluid advice, especially with diarrhoea or heart or kidney disease. Adjust activity and rest to your energy. Report dizziness, breathlessness or fatigue that limits daily tasks so treatable causes can be checked.

Emotions, relationships and quality of life

Ask for help with mood, sleep, relationships, intimacy or caring responsibilities when you need it. Symptom and palliative-care teams can support comfort and quality of life while anticancer treatment continues.

Practical help

Ask your specialist nurse or social worker about transport, employment, finances and help at home. Tell them about caring responsibilities and ask which support organisations fit your needs.

A record that works without a phone

Keep a paper schedule, current medicine list and real contact numbers together. The handbook can be read without internet access. Use the written emergency plan whenever you need urgent help.

Questions to take to the next visit

For the next visit, choose the questions that matter most: my exact diagnosis; the aim of treatment; which tests I need and why; who explains results; when to call; and what help is available.

Questions worth asking

Does my receptor scan and exact NET diagnosis meet this PRRT indication?

The specialist team must confirm progressive, well-differentiated G1/G2 SSTR-positive gastroenteropancreatic neuroendocrine tumour (GEP-NET) and assess other eligibility factors. A receptor scan alone does not establish that treatment is suitable.

Will the first or later visits require overnight admission?

The treatment information describes an overnight stay for the first treatment; later treatments may be day case or overnight. Confirm the actual admission and discharge plan with the PRRT service.

Who provides the written stop/restart dates for my SSA injections?

The referring NET consultant and nuclear-medicine team coordinate the injection plan. Get exact written dates; do not stop or restart an SSA on your own.

When are my fortnightly and post-course blood tests?

NCCP specifies two-weekly bloods between administrations and continued tests after the course. Your specialist team may arrange additional early post-course checks. Ask for one reconciled written schedule and result-review contact.

Which household, work and travel precautions apply, and on what dates do they end?

The medical physicist gives individual restrictions, which can extend to three weeks for some contacts. Obtain each end date and discuss transport, sleeping, childcare and work before admission.

What symptoms suggest a hormone crisis, and who do I call after discharge?

Sudden severe flushing, faintness, diarrhoea, wheeze or breathing trouble can require urgent assessment, particularly in the first 48 hours. Follow the service's written emergency plan and do not delay urgent care because of radiation precautions.

What fertility, contraception and long-term follow-up plan do I need?

Discuss fertility before starting. Current EMA/HSE information requires contraception during treatment and for seven months afterwards for patients who could become pregnant, and four months for men with partners who could become pregnant. Blood and organ-function follow-up continues after the fourth administration.

> **Try explaining the plan back:** Can you explain the four planned visits, your blood-test timetable, the separate SSA injection plan, radiation precautions and who to call for sudden hormone symptoms?

Specialist receptor-targeted radiation treatment This guide covers the Irish NCCP G1/G2 somatostatin receptor (SSTR)-positive gastroenteropancreatic neuroendocrine tumour (GEP-NET) regimen. A receptor scan, specialist eligibility review and individual radiation-safety plan are required.

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

PRRT administration with kidney protection

Active medicines: Lutetium-177 oxodotreotide + Lysine / arginine amino acids

Emetogenic risk: Moderate

Medicine	Dose / route	When
5-HT3 antagonist antiemetic	Agent and dose prescribed by the nuclear-medicine service	At least 30 minutes before the amino-acid infusion NCCP regimen recommendation; additional antiemetics if needed under local policy.

Medicine	Dose / route	When
Lysine / arginine amino-acid infusion	Formulation and volume prescribed by the specialist service	Start 30 minutes before Lutathera and continue for four hours Required kidney protection. Check potassium beforehand; fluid/renal/cardiac risk may require an individual plan.

Additional phase-specific support

- A lysine/arginine infusion is required for kidney protection and begins before Lutathera. Follow the service's fluid and toileting advice. Tell staff about heart or kidney problems; do not use a generic fluid target.
- PRRT has moderate emetogenic potential. A 5-HT3 anti-sickness medicine is given at least 30 minutes before the amino-acid infusion, with additional treatment if needed. Tell staff promptly if nausea starts.
- Long-acting somatostatin analogue (SSA) injections can interfere with receptor targeting. NCCP uses a 30-day interval before PRRT and short-acting injections are withheld for 24 hours. The referring neuroendocrine tumour (NET) consultant decides your exact stop and restart plan, balancing hormone control. Never stop injections on your own.
- The medical physicist gives individual written and verbal advice. Contact with young children or pregnant people may need restriction for up to three weeks, and close-contact work may require up to two weeks away. Arrange transport, sleeping, toileting, laundry and caring responsibilities before admission.
- Radiation safety: individual medical-physicist instructions and official radiation card; the SACTCheck wallet card does not replace them.
- The long treatment information describes earlier additional post-course blood tests and service scan timings; reconcile the written service plan with the NCCP minimum monitoring sequence.
- Do not restart SSA injections, prescribe steroids or add electrolyte/fluid treatment without the relevant specialist plan.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

What to watch for

You do not need to know which medicine caused a symptom before calling.

1 Energy / wellbeing

PRRT

Tiredness may last beyond the treatment day. Severe weakness, confusion, seizure or collapse needs urgent assessment.

3 Heart / fluids / potassium

Amino acids / hormone effects

Potassium or fluid changes can affect the heart. Report palpitations, new swelling or breathlessness promptly; severe symptoms are an emergency.

5 Liver

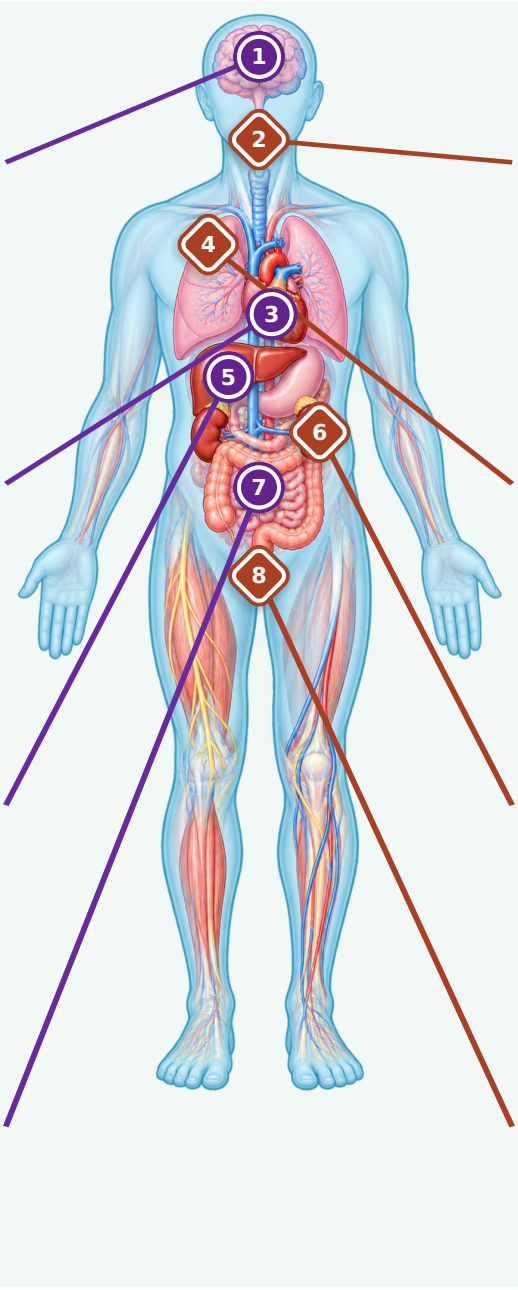
PRRT

Liver tests are monitored before, between and after treatments. Report yellow skin or eyes, dark urine or persistent severe abdominal pain.

7 Bowel / hormone symptoms

PRRT / NET

Abdominal pain, diarrhoea or sudden hormone release can occur. Severe flushing, faintness, diarrhoea or wheeze needs urgent advice.



2 Appetite / sickness

PRRT / amino acids

Nausea and vomiting can occur, especially around treatment. Tell staff early and follow the anti-sickness and fluid plan.

4 Bone marrow / blood

PRRT

Counts may fall during or after the course. Report fever, bruising or bleeding. Rare late marrow disorders require ongoing follow-up.

6 Kidneys / urine

PRRT

Amino acids reduce kidney exposure but do not remove risk. Keep renal blood tests and report much less urine or dehydration.

8 Fertility / pregnancy

PRRT

Discuss fertility before treatment. Avoid pregnancy using the written contraception plan; follow individual radiation precautions for close contacts.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Seek urgent help for sudden severe flushing with faintness, severe diarrhoea, wheeze or breathing trouble, especially in the first 48 hours; follow the hormone-crisis plan. Fever, bleeding, much less urine, persistent severe abdominal pain, muscle weakness or confusion needs prompt advice. Tell emergency staff when you received PRRT; radiation precautions must not delay urgent care.

When to get help

Call 112 / 999 for severe breathing difficulty, severe chest pain, collapse or a seizure.

Call oncology immediately: temperature 37.5°C or above, 36°C or below, shivering or suddenly feeling unwell, even with a normal temperature. Follow your hospital's temperature instructions. If you cannot reach the team, seek urgent medical assessment. Do not wait for the next appointment.

Oncology daytime: 087 12345678 · **Out of hours:** 087 87654321

EXAMPLE ONLY — team must replace both numbers before patient use.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **Use the prescribed local supportive-care plan; independent clinical/pharmacy review pending**
   https://healthservice.hse.ie/documents/6683/642_v2_Lutetium_Lu177_oxodotretide.pdf
2. **EMA Lutathera product information and patient leaflet · updated 14 August 2026 ↗**
   https://www.ema.europa.eu/en/documents/product-information/lutathera-epar-product-information_en.pdf
3. **HSE / PRRT service: introductory patient information ↗**
   https://assets.hse.ie/media/documents/Neuroendocrine_PRRT_Patient_Information_Leaflet_short.docx
4. **NCI PDQ: gastrointestinal neuroendocrine tumours (NETs) · selected marker background**
   https://www.cancer.gov/types/gi-neuroendocrine-tumors/hp/gi-neuroendocrine-treatment-pdq
5. **NCI PDQ: pancreatic NETs · hormone and digestive background**
   https://www.cancer.gov/types/pancreatic/hp/pnet-treatment-pdq
6. **NETTER-1 final overall-survival and long-term safety report**
   https://pubmed.ncbi.nlm.nih.gov/34793718/
7. **HSE / PRRT service: detailed patient information**
   https://assets.hse.ie/media/documents/Neuroendocrine_PRRT_Patient_Information_Leaflet_long.docx
8. **NCI: fatigue during cancer treatment**
   https://www.cancer.gov/about-cancer/treatment/side-effects/fatigue
9. **NANETS expert consensus: high-grade neuroendocrine neoplasms, 2023**
   https://pmc.ncbi.nlm.nih.gov/articles/PMC10388681/
10. **NCI: gastrointestinal neuroendocrine tumours — patient information**
    https://www.cancer.gov/types/gi-neuroendocrine-tumors/patient/gi-neuroendocrine-treatment-pdq
11. **NCI: eating and nutrition during cancer treatment**
    https://www.cancer.gov/about-cancer/treatment/side-effects/nutrition
12. **NCI: palliative care during cancer treatment**
    https://www.cancer.gov/about-cancer/advanced-cancer/care-choices/palliative-care-fact-sheet
13. **NCI: pancreatic neuroendocrine tumour diagnosis and tests**
    https://www.cancer.gov/types/pancreatic/neuroendocrine-tumors/diagnosis
14. **NCI: pancreatic neuroendocrine tumour symptoms**
    https://www.cancer.gov/types/pancreatic/neuroendocrine-tumors/signs-symptoms
15. **NCI: pancreatic neuroendocrine tumour treatment**
    https://www.cancer.gov/types/pancreatic/neuroendocrine-tumors/treatment
16. **NCI: pancreatic neuroendocrine tumour types**
    https://www.cancer.gov/types/pancreatic/neuroendocrine-tumors